

Module 08: Virtual Memory and Demand Paging

AKTU BCS401

Unit 4

Virtual Memory Principles

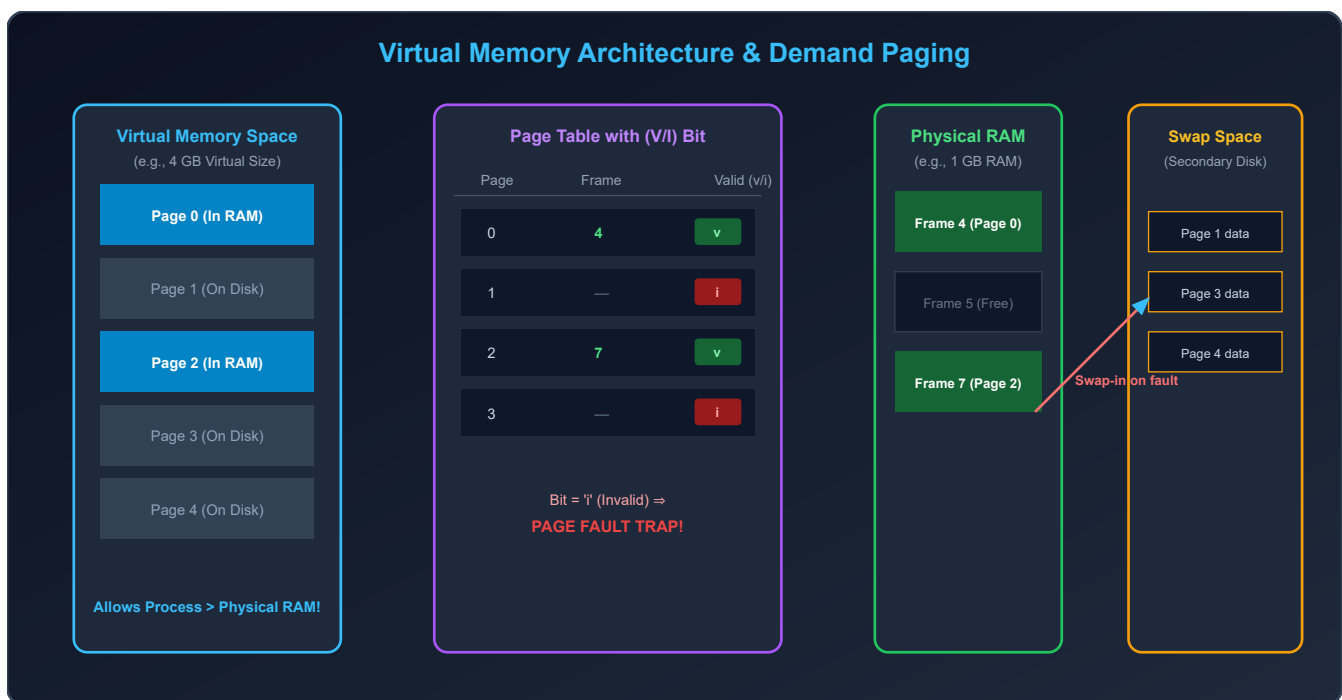
1. Virtual Memory Concept

Virtual memory user logical address space ko physical memory se alag karta hai, jisse process size physical RAM se bada ho sakta hai aur multiprogramming degree dramatically badh jaati hai.

2. Demand Paging & Valid-Invalid (V/I) Bit

Pages ko memory me tabhi load kiya jata hai jab unki demand ho (Lazy Swapper). Page Table me har entry me ek bit hota hai:

- **Valid (v):** Page RAM ke frame me present hai. Address translation direct hoti hai.
- **Invalid (i):** Page RAM me nahi hai balki disk par hai. MMU **Page Fault Trap** trigger karta hai.



3. Hardware Requirements

1. **Page table with V/I bit**
2. **Swap Space (Backing store)** on high-speed disk.
3. **Instruction Restartability:** Fault resolve hone ke baad trapped instruction ko exact usi state se restart karne ki capability.